NHI NGUYEN

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INTERESTS

Explanation faithfulness, Metacognition in LLMs and agents, Trustworthy AI evaluation, Causal inference

EDUCATION

Courant Institute, New York University — New York, NY
PhD Candidate in Computer Science — *Aug 2023 - Present*
Advisor: Rajesh Ranganath

Yale University — New Haven, CT
BS/MS in Computer Science, BS in Applied Mathematics, Summa Cum Laude — *Aug 2019 - May 2023*
Honors in Computer Science, Honors in Applied Mathematics

PUBLICATIONS

Conference

- Code as Explanations: Executable and Verifiable LLM Explanations. **Nhi Nguyen***, Shauli Ravfogel, Rajesh Ranganath. *NeurIPS (2026)*
- What LLMs Explain Is Not What They Believe: Evaluating Explanation Sufficiency Under Models' Own Input Beliefs. **Nhi Nguyen***, Shauli Ravfogel, Rajesh Ranganath. *ICML (2026)*
- FIMP: Foundation Model-Informed Message Passing for Graph Neural Networks. Syed Rizvi*, Nazreen Pallikkavaliyaveetil*, David Zhang, Zhuoyang Lyu, **Nhi Nguyen**, Harry Lyu, Benjamin Christensen, Josue Caro, Antonio Fonseca, Emanuele Zappala, Maryam Bagherian, Christopher Averill, Chadi Abdallah, Amin Karbasi, Rex Ying, Maria Brbic, Rahul Dhodapkar, David van Dijk. *TMLR (2026)*
- Explanations that reveal all through the definition of encoding. Aahlad Puli*, **Nhi Nguyen***, Rajesh Ranganath. *NeurIPS (2024)*

Workshop/Preprint

- Black Box Causal Inference: Effect Estimation via Meta Prediction. Lucius EJ Bynum, Aahlad Manas Puli, Diego Herrero-Quevedo, **Nhi Nguyen**, Carlos Fernandez-Granda, Kyunghyun Cho, Rajesh Ranganath. *arXiv:2503.05985 (2025)*
- Do LLMs Lie About What They Use? Benchmark for Metacognitive Truthfulness in Large Language Models. **Nhi Nguyen***, Shauli Ravfogel, Rajesh Ranganath. *Actionable Interpretability Workshop ICML (2025)*
- A Flow Artist for Embedding High -Dimensional Cellular Data within Vector Fields. Kincaid MacDonald*, Dhananjay Bhaskar*, Guy Thampakkul, **Nhi Nguyen**, Joia Zhang, Tesfa Asmara, Michael Perlmutter, Ian Adelstein, Smita Krishnaswamy. *MLSP (2023)*

RESEARCH EXPERIENCE

New York University, Rajesh Lab — New York, NY
PhD Researcher — *Aug 2023 - Present*

- Formalized key properties of LLM explanation faithfulness using information-theoretic frameworks, proving formal guarantees and validating them across 4 families of open-sourced LLMs
- Developed scalable, automatic metrics (STRIPE-X, SCSuff, Consistency) to evaluate explanation transparency, sufficiency, and alignment with model behavior
- Proposed executable code explanations as a new explanation paradigm and built an automated generation and verification pipeline, achieving 10x - 170x shorter explanations than chain-of-thought
- Implemented and open-sourced LLM evaluation pipelines for measuring explanation sufficiency using alternative inputs the model itself considers likely

Capital One, First Party Fraud Team McLean, VA
Data Science Intern *Jun 2026 - Aug 2026*

- Developed and scaled a graph-based fraud model using Label Propagation and edge weighting to detect organized rings 3 months in advance, mitigating $71M in risk and $474K in losses
- Designed and implemented a modular pipeline and 3-tier evaluation framework (community, statistical, business metrics) to accelerate iteration and uncover new fraud signals from customer connections

Character AI, Post-training Team Redwood City, CA
Research Intern *Jun 2025 - Aug 2025*

- Trained an offline reward model on user preference data over entertainment chat model responses, achieving a 15% AUROC improvement to support reinforcement learning and inference-time reranking

VinAI, Residency Program Hanoi, Viet Nam
Research Intern *Jun 2023 - Aug 2023*

- Engineered a real-time drunk-driving detection model using features derived from in-car cameras and sensors, achieving 80% accuracy

Van Dijk Lab at Yale New Haven, CT
Undergraduate Thesis: Attention as Message Passing for Graph Neural Networks *Sep 2022 - May 2023*
Advisor: David van Dijk

- Constructed a novel GNN layer using multi-headed attention for interpretable feature interactions and a synthetic cyclic simulation dataset to evaluate interaction recovery; Resulted in preprint AMPNet

Summer Undergraduate Math Research at Yale New Haven, CT
Project: Embedding High-Dimensional Cellular Data within Vector Fields *Jun 2022 - Sep 2022*
Advisor: Smita Krishnaswamy

- Designed a neural network that jointly embeds high-dimensional single-cell RNA velocity data and its vector field, supporting the analysis and visualization of biological dynamics; Published at MLSP 2023

Tietjen Lab, Team Next Gen Research Assistantship New Haven, CT
Project: Cluster protein activities in donated kidneys *Nov 2021 - Jun 2022*
Advisor: Gregory Tietjen

- Investigated and compared methods for clustering proteins by shared Gene Ontology annotations and expression activity in donated kidneys, supporting downstream assessment of graft viability

AWARDS & FELLOWSHIPS

Top Reviewer at NeurIPS 2026, ICML 2026 2026
DeepMind Scholarship 2023
MacCracken Doctoral Fellowship, NYU Graduate School of Arts and Sciences 2023
Russell Henry Chittenden Prize for senior majoring in the natural sciences 2023
Phi Beta Kappa Junior Inductee 2022

SKILLS

Programming & ML: Python, PyTorch, Hugging Face Transformers, Scikit-learn, Pandas, Spark
Infrastructure & Tools: HPC/SLURM, Git, GitHub, Weights & Biases, Multi-GPU training
Research methods: LLM evaluation, Statistical analysis, Model interpretability, Causal inference

ACADEMIC SERVICE

Reviewer: NeurIPS 2025-2026, ICLR 2026-2027, ICML 2026, AAAI 2026, TMLR 2026 *2025 - 2026*
Grad TA at NYU: Machine Learning *SP 2026*
TA at Yale: Deep Learning Theory & Applications; Natural Language Processing; *2020 - 2023*
Stochastic Processes; Optimization & Computation; Data Exploration & Analysis; Probability & Statistics